

TalentFlow: Multi-Source Candidate Data Transformer

1. Pipeline Architecture

The system is a deterministic, modular 6-stage DAG to ensure traceability and strict error isolation:

- **Ingest (Detect):** Discovers files in deterministic alphabetical order, validates MIME types, and routes to appropriate parsers.
- **Extract:** Source-specific parsers extract intermediate data. The ResumeParser uses strict regex and structural cues, prioritizing "honestly-empty" over "wrong-but-confident" by extracting only from designated sections.
- **Normalize:** Cleans extracted data (e.g., standardizing phones, mapping countries, resolving skill aliases).
- **Merge:** Resolves identity and merges data across sources using a Union-Find algorithm.
- **Score Confidence:** Evaluates data completeness, source reliability, and cross-source agreement.
- **Project-to-Output (Validate):** Reshapes the canonical internal record into a user-defined JSON schema.

2. Canonical Schema & Normalizations

The internal canonical profile provides a strict structure:

- **Core Fields:** `candidate_id` (UUID hash), `full_name` (String), `emails` (String Array), `links` (Object).
- **Phones** (String Array): Normalized via the `phonenumbers` library to strict E.164 format. Invalid sequences drop to null.
- **Location** (Object): `{city, region, country}`. Country normalized by mapping strings ("United Kingdom") to ISO-3166 alpha-2 ("GB") using `pycountry`.
- **Experience, Education:** Dates normalized to strict YYYY-MM format using fuzzy parsing with `dateutil`.
- **Skills** (Array): Normalized against a canonical alias dictionary (e.g., "js" → "JavaScript", "ML" → "Machine Learning").
- **Provenance** (Array): Tracks lineage `{field, source, method}` for every canonical data point.

3. Merge / Conflict-Resolution Policy & Confidence

- **Identity Resolution:** Identity is resolved using a Union-Find graph primarily via email matching (with exact name match as fallback).
- **Conflict Policy:** When merging, sources are ranked by structural trustworthiness: JSON [0.9] > CSV [0.7] > PDF [0.65] > GitHub [0.4]. If sources provide conflicting scalar values (e.g., headline), the highest-weight source wins. Array fields (like skills) are unioned and deduplicated.
- **Confidence Formula:**
 - Per-field confidence = $\text{source_weight} \times \text{completeness} \times \text{agreement_bonus}$
 - Completeness = 0.0 if field is null/empty, 1.0 otherwise.
 - Agreement bonus (1.2×) applied if 2+ independent sources agree on exact value, reflecting higher trustworthiness when sources corroborate.
 - Overall confidence = arithmetic mean of all field confidences.

4. Runtime Configurable Output (Projection Layer)

TalentFlow decouples its internal state from the output shape using a dynamic JSON Projection Config:

- **Selection & Remapping:** Resolves nested paths via array indexing ("from": "emails[0]" maps to `primary_email`).
- **Format Toggles:** Applies per-field normalizations directly at output (e.g., "normalize": "E164") and toggles features like `include_provenance` and `include_confidence`.
- **On-Missing Policy:** Defines fallback behavior if required data is missing: "null" sets to null, "omit" removes the key entirely, "error" aborts pipeline with `ProjectionError`.

5. Edge Cases & Scope Exclusions

- **Edge Cases Successfully Handled:**
 - **Garbage Documents (e.g., Financial Invoices):** Strict regex guardrails (requiring headers like "Technical Skills") prevent extracting random fragments. If unsure, the parser yields empty lists rather than false positives.
 - **GitHub API Rate Limits:** If a 403 Rate Limit is encountered, the parser logs a warning and returns partial data from the basic API call without crashing the pipeline.
 - **Array Access on Empty Arrays:** If user config requests `emails[0]` but no emails were found, the projection layer gracefully applies the `on_missing` policy (returns null if policy is "null", omits the field if "omit", or throws `ProjectionError` if "error").
- **Deliberately Descoped (Due to Time/Constraints):**
 - **Image-only PDF OCR:** Excluded heavy dependencies like Tesseract OCR. Pure image PDFs currently return an empty record with very low confidence.
 - **LLM Extraction:** Opted for deterministic regex over generative AI to guarantee zero-latency execution, perfect determinism, and 100% explainability without network dependencies or API costs.
 - **LinkedIn Profile Parsing:** Completely removed due to LinkedIn's anti-scraping policies and lack of public API access without paid enterprise license.